

Follicle-stimulating Hormone (FSH)

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Excessive FSH and LH are found in hypogonadism, anorchia, gonadal failure,¹ complete testicular feminization syndrome, menopause, Klinefelter syndrome, alcoholism, and castration. FSH and LH are pituitary products, useful to distinguish primary gonadal failure from secondary (hypothalamic/pituitary) causes of gonadal failure, menstrual disturbances, and amenorrhea. Useful in defining menstrual cycle phases in infertility evaluation of women and testicular dysfunction in men. FSH is commonly used with LH, which also is a gonadotropin. Both are low in pituitary or hypothalamic failure. FSH and LH levels are high following menopause.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

Secretion of both LH and FSH are pulsatile, in response to the normal intermittent release of gonadotropin-releasing hormone (GnRH). In addition, in females, both FSH and LH vary over the course of the menstrual cycle, with peaks at time of ovulation. Thus, interpretation of a single determination may be difficult. It has been suggested that samples be obtained at 15- to 30-minute intervals and equal volumes of serum be pooled to decrease the effect of pulsatile secretion.

As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or who have received them for diagnostic purposes.² In rare cases, interference due to extremely high titers of antibodies to streptavidin and ruthenium can occur.² The test contains additives, which minimize these effects.

Custom Additional Information

FSH is a glycoprotein consisting of two subunits (α - and β -chains). Its molecular weight is approximately 32,000 daltons. FSH together with LH (luteinizing hormone), belongs to the gonadotropin family. FSH and LH regulate and stimulate the growth and function of the gonads (ovaries and testes) synergistically.³

FSH and LH are released in pulses from the gonadotropic cells of the anterior pituitary. The levels of the circulating hormones are controlled by steroid hormones via negative feedback to the hypothalamus. In the ovaries, FSH, together with LH, stimulates the growth and maturation of the follicle and hence also the biosynthesis of estrogens in the follicles.

In women, the gonadotropins act within the hypothalamus-pituitary-ovary regulating circuit to control the menstrual cycle.^{1,4} The FSH level shows a peak at midcycle, although this is less marked than with LH. Due to changes in ovarian function and reduced estrogen secretion, high FSH concentrations occur during menopause.¹ The determination of FSH in conjunction with LH is utilized for the following indications: congenital diseases with chromosome aberrations, polycystic ovaries (PCO), amenorrhea (causes), and menopausal syndrome.

In men, FSH serves to induce spermatogonium development. Determination of the FSH concentration is used in the elucidation of dysfunctions within the hypothalamus-pituitary-gonads system. Depressed gonadotropin levels in men occur in azoospermia.^{1,3,5,6}

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

If a red-top tube or plasma is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Children (Male and Female) (mIU/mL)		
<24 h	<0.2–0.8	
1 d	<0.2–0.8	
2 d	<0.2–0.8	
3 d	<0.2–2.4	
4 d	<0.2–2.3	
5 d	<0.2–3.4	
6 d	<0.2–4.5	
7 d	<0.2–21.4	
8 to 30 d	<0.2–22.2	
Age	Children (Male)	Children (Female)
1 to 12 m	Not established	Not established
1 to 4 y	0.2–2.8	0.2–11.1
5 to 9 y	0.4–3.8	0.3–11.1
10 to 12 y	0.4–4.6	2.1–11.1
13 to 16 y	1.5–12.9	1.6–17.0
Adult Male (mIU/mL): 1.5–12.4		

Children (Male and Female) (mIU/mL)	
Adult Female (mIU/mL)	
follicular	3.5–12.5
ovulatory	4.7–21.5
luteal	1.7–7.7
postmenopausal	25.8–134.8

Storage Instructions

Room temperature

References

Jaakkola T, Ding YQ, Kellokumpu-Lehtinen P, et al. The ratios of serum bioactive/immunoreactive luteinizing hormone and follicle-stimulating hormone in various clinical conditions with increased and decreased gonadotropin secretion: Re-evaluation by a highly sensitive immunometric assay. *J Clin Endocrinol Metab*. 1990 Jun; 70(6):1496-1505.

PubMed 2140831 (<http://www.ncbi.nlm.nih.gov/pubmed/2140831>)

Layman LC, Wilson JT, Huey LO, Lanclos KD, Plouffe L Jr, McDonough PG. Gonadotropin-releasing hormone, follicle-stimulating hormone beta, luteinizing hormone beta gene structure in idiopathic hypogonadotropic hypogonadism. *Fertil Steril*. 1992 Jan; 57(1):42-49. PubMed 1730329

(<http://www.ncbi.nlm.nih.gov/pubmed/1730329>)

Footnotes

1. Runnebaum B, Rabe T. *Gynäkologische Endokrinologie und Fortpflanzungsmedizin*. Berlin, Germany: Springer Verlag;1994. Band 1:17, 253-255; Band 2: 152-154, 360, 348. ISBN 3-540-57345-3, ISBN 3-540-57347-X.

2. FSH on Elecsys 1010/2010 and Modular Analytics E170, package insert 2007-10, V 14, Indianapolis, Ind: Roche Diagnostics; 2007.
3. Johnson MR, Carter G, Grint C, Lightman SL. Relationship between ovarian steroids, gonadotropins and relaxin during the menstrual cycle. *Acta Endocrinol (Copenh)*. 1993 Aug; 129(2):121-125. PubMed 8372595 (<http://www.ncbi.nlm.nih.gov/pubmed/8372595>)
4. Beastall GH, Ferguson KM, O'Reilly DS, Seth J, Sheridan B. Assays for follicle stimulating hormone and luteinizing hormone: Guidelines for the provision of a clinical biochemistry service. *Ann Clin Biochem*. 1987 May; 24(Pt 3):246-262. PubMed 3111341 (<http://www.ncbi.nlm.nih.gov/pubmed/3111341>)
5. Schmidt-Mathiesen H. *Gynäkologie und Geburtshilfe*. Schattauer Verlag; 1992.
6. Scott MG, Ladenson JH, Green ED, Gast MJ. Hormonal evaluation of female infertility and reproductive disorders. *Clin Chem*. 1989 Apr; 35(4):620-630. PubMed 2522836 (<http://www.ncbi.nlm.nih.gov/pubmed/2522836>)